

The empirical recurrence for OEIS A253616, recovered from its tail

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8 September 2026

Abstract

OEIS A253616 records “Empirical recurrence of order 80” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 114$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 82$. Its last coefficient is $c_{80} = 9 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A253616 is “Number of $(1+2)X(n+2)$ 0..2 arrays with every consecutive three elements in every row and diagonal having one or two distinct values, and in every column and antidiagonal having two or three distinct values, and new values 0 upwards introduced in row major order.”. It has offset 1 and begins

762, 6916, 61634, 551335, 4928315, 44053769, 393764260, 3519639788, ...

The entry states:

Empirical recurrence of order 80 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:08:12, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 114$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 114$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 228$ exact terms from $n = 2$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{80} c_i a(n-i),$$

c_1	11	c_{21}	44759245	c_{41}	8116494639	c_{61}	−52575654
c_2	−12	c_{22}	579259742	c_{42}	−13204995161	c_{62}	−111093754
c_3	−8	c_{23}	847747636	c_{43}	−6161063367	c_{63}	−81123275
c_4	−498	c_{24}	−720359265	c_{44}	29854605339	c_{64}	−13153632
c_5	87	c_{25}	−3314219695	c_{45}	13860828268	c_{65}	75380937
c_6	2563	c_{26}	−826319393	c_{46}	−37931502268	c_{66}	26812042
c_7	10208	c_{27}	5880965496	c_{47}	−13206028636	c_{67}	−17575744
c_8	6958	c_{28}	4710822630	c_{48}	20066211986	c_{68}	−11129888
c_9	−53376	c_{29}	−3872308251	c_{49}	9561188300	c_{69}	−180087
c_{10}	−128016	c_{30}	−9671101203	c_{50}	1118558074	c_{70}	2246792
c_{11}	−166899	c_{31}	−4976211518	c_{51}	−9534300460	c_{71}	902633
c_{12}	209290	c_{32}	7944641565	c_{52}	−6050033126	c_{72}	−207832
c_{13}	692735	c_{33}	14861837791	c_{53}	7266228096	c_{73}	−209209
c_{14}	1810917	c_{34}	5551371851	c_{54}	3392855506	c_{74}	−34104
c_{15}	3619276	c_{35}	−25073405604	c_{55}	−1401496838	c_{75}	23779
c_{16}	1927564	c_{36}	−17478450572	c_{56}	−920816650	c_{76}	9745
c_{17}	−6669942	c_{37}	33570570207	c_{57}	−910531323	c_{77}	−1708
c_{18}	−29897568	c_{38}	11126322145	c_{58}	−166069052	c_{78}	−248
c_{19}	−54574776	c_{39}	−25662177923	c_{59}	381577784	c_{79}	67
c_{20}	−83014214	c_{40}	3224251206	c_{60}	268238782	c_{80}	9

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 82$, and it is the recurrence OEIS A253616 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1x^{80-1} - \dots - c_{80}$. Its constant term is $-c_{80} = -9$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 16 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 9.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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